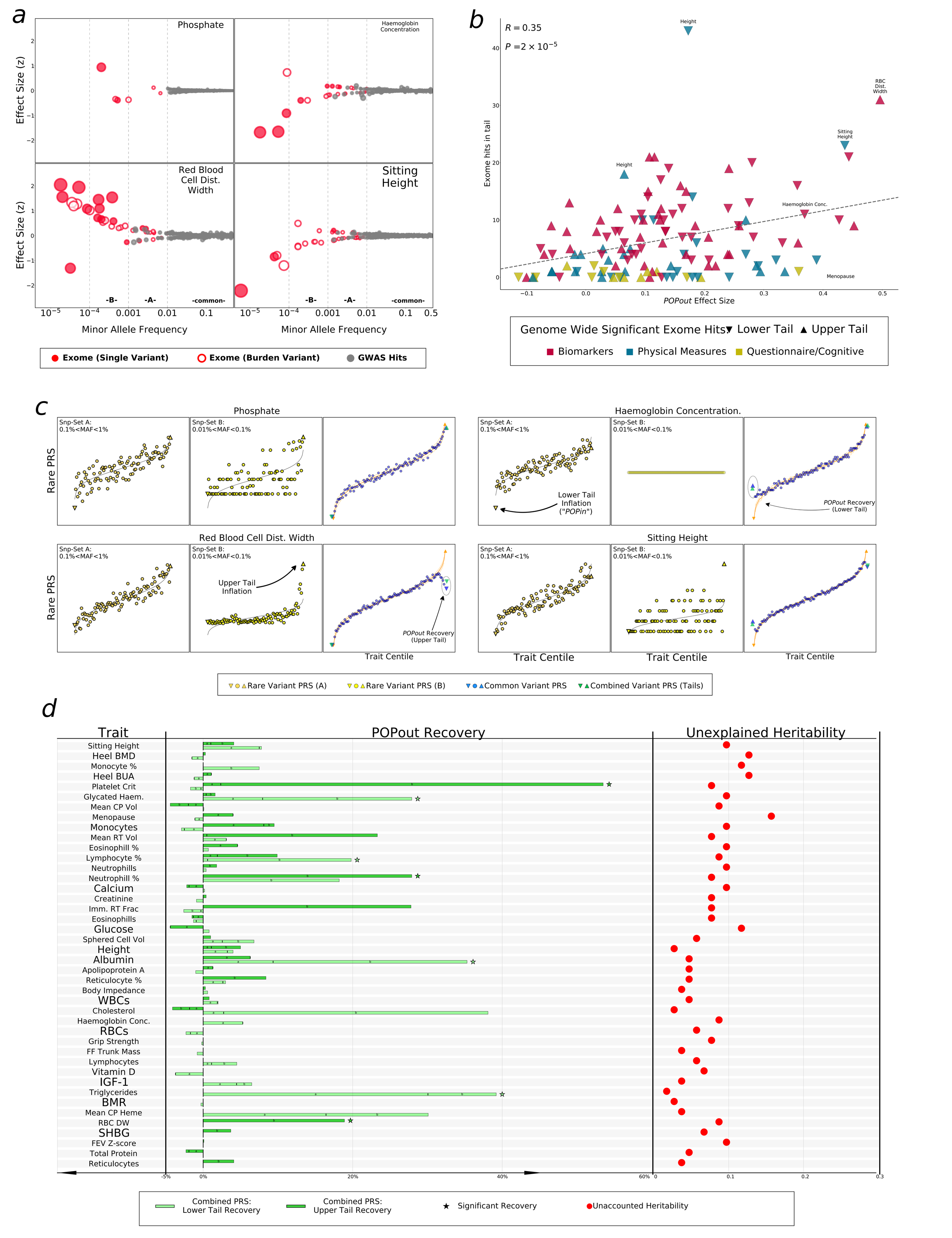
a
Phosphate
Haemoglobin Concentration
Red Blood Cell Dist. Width
Sitting Height
Effect Size (z)
-B-
-A-
-common-
Minor Allele Frequency
Exome (Single Variant)
Exome (Burden Variant)
GWAS Hits
b
R = 0.35
P = 2 × 10^−5
Height
RBC Dist. Width
Sitting Height
Haemoglobin Conc.
Menopause
Exome hits in tail
POPout Effect Size
Genome Wide Significant Exome Hits ▼ Lower Tail ▲ Upper Tail
Biomarkers
Physical Measures
Questionnaire/Cognitive
c
Phosphate
Haemoglobin Concentration.
Red Blood Cell Dist. Width
Sitting Height
Snp-Set A: 0.1%<MAF<1%
Snp-Set B: 0.01%<MAF<0.1%
Rare PRS
Lower Tail Inflation ("POPin")
POPout Recovery (Lower Tail)
Upper Tail Inflation
POPout Recovery (Upper Tail)
Trait Centile
Rare Variant PRS (A)
Rare Variant PRS (B)
Common Variant PRS
Combined Variant PRS (Tails)
d
Trait
POPout Recovery
Unexplained Heritability
Sitting Height
Heel BMD
Monocyte %
Heel BUA
Platelet Crit
Glycated Haem.
Mean CP Vol
Menopause
Monocytes
Mean RT Vol
Eosinophill %
Lymphocyte %
Neutrophills
Neutrophill %
Calcium
Creatinine
Imm. RT Frac
Eosinophills
Glucose
Sphered Cell Vol
Height
Albumin
Apolipoprotein A
Reticulocyte %
Body Impedance
WBCs
Cholesterol
Haemoglobin Conc.
RBCs
Grip Strength
FF Trunk Mass
Lymphocytes
Vitamin D
IGF-1
Triglycerides
BMR
Mean CP Heme
RBC DW
SHBG
FEV Z-score
Total Protein
Reticulocytes
-5%
0%
20%
40%
60%
0
0.1
0.2
0.3
Combined PRS: Lower Tail Recovery
Combined PRS: Upper Tail Recovery
Significant Recovery
Unaccounted Heritability